

Vector product or cross product

→ Let $\vec{a} = (a_1, a_2, a_3)$ and $\vec{b} = (b_1, b_2, b_3)$ be two vectors. Then their vector product or cross product is denoted by $\vec{a} \times \vec{b}$ and is defined by

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$$= \vec{i}(a_2 b_3 - a_3 b_2) - \vec{j}(a_1 b_3 - a_3 b_1) + \vec{k}(a_1 b_2 - a_2 b_1)$$

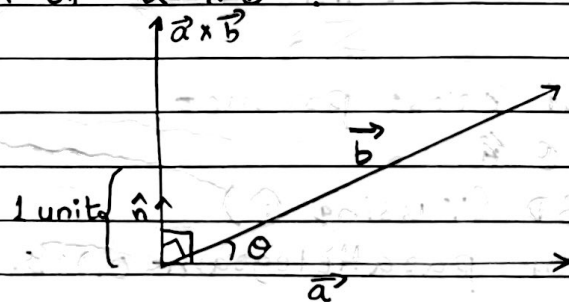
$$= (a_2 b_3 - a_3 b_2, a_3 b_1 - a_1 b_3, a_1 b_2 - a_2 b_1)$$

• Let $\vec{a}$ and $\vec{b}$ be two vectors and θ be the angle between them. Then their vector product or cross product is denoted by $\vec{a} \times \vec{b}$ and is defined by

$$\vec{a} \times \vec{b} = ab \sin \theta \hat{n}$$

Where,

$\hat{n}$ is a unit vector perpendicular to both $\vec{a}$ and $\vec{b}$ which gives the direction of $\vec{a} \times \vec{b}$.



Note: ① If θ be the angle between $\vec{a}$ and $\vec{b}$, then

$$|\vec{a} \times \vec{b}| = |\vec{a}| |\vec{b}| \sin \theta$$

$$\sin \theta = \frac{|\vec{a} \times \vec{b}|}{|\vec{a}| |\vec{b}|}$$

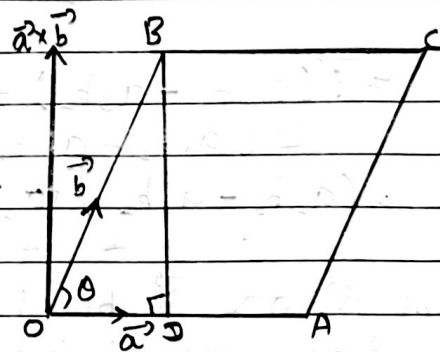
② The vector perpendicular to both $\vec{a}$ and $\vec{b}$ is $\vec{a} \times \vec{b}$

③ The unit vector perpendicular to both $\vec{a}$ and $\vec{b}$ is $\frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|}$

Geometrical Interpretation of cross product

Let θ be the angle between $\vec{a}$ and $\vec{b}$.

Suppose the magnitude and direction of $\vec{a}$ and $\vec{b}$ are respectively represented by two adjacent sides OA and OB of a parallelogram OACB.



Draw $BD \perp OA$.

In right angled $\triangle ODB$

$$\sin \theta = \frac{BD}{OB} \Rightarrow \sin \theta = \frac{BD}{b}$$

$$\Rightarrow BD = b \sin \theta \rightarrow (1)$$

By definition of cross product,

$$|\vec{a} \times \vec{b}| = ab \sin \theta$$

$$= OA \times BD \quad (\because \text{using (1)})$$

$$= \text{Area of parallelogram OACB.}$$

Hence, the magnitude of cross product of two vectors $\vec{a}$ and $\vec{b}$ is equal to the area of parallelogram whose two adjacent sides are ~~vectors~~ $\vec{a}$ and $\vec{b}$.

Notes: The Area of parallelogram whose two adjacent sides are $\vec{a}$ & $\vec{b}$ is $|\vec{a} \times \vec{b}|$.

: The Area of triangle whose two adjacent sides are $\vec{a}$ & $\vec{b}$ is $\frac{1}{2} |\vec{a} \times \vec{b}|$

: The Area of parallelogram whose two diagonals are d_1 and d_2 is $\frac{1}{2} |d_1 \times d_2|$

Exercise 10.2

Part B: Creative Questions

2) If $\vec{a} = 3\vec{i} + 4\vec{j} - 5\vec{k}$ and $\vec{b} = 7\vec{i} - 3\vec{j} + 6\vec{k}$, calculate $(\vec{a} + \vec{b}) \times (\vec{a} - \vec{b})$ and $|\vec{a} + \vec{b}| \times |\vec{a} - \vec{b}|$.

→ Soln,

$$\vec{a} + \vec{b} = 3\vec{i} + 4\vec{j} - 5\vec{k} + 7\vec{i} - 3\vec{j} + 6\vec{k} \\ = 10\vec{i} + \vec{j} + \vec{k}$$

$$\vec{a} - \vec{b} = 3\vec{i} + 4\vec{j} - 5\vec{k} - 7\vec{i} + 3\vec{j} - 6\vec{k} \\ = -4\vec{i} + 7\vec{j} - 11\vec{k}$$

Now,

$$(\vec{a} + \vec{b}) \times (\vec{a} - \vec{b}) = \vec{a} \times \vec{b} \\ = (10\vec{i} + \vec{j} + \vec{k}) \times (-4\vec{i} + 7\vec{j} - 11\vec{k}) \\ = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 10 & 1 & 1 \\ -4 & 7 & -11 \end{vmatrix}$$

$$= (-11 - 7)\vec{i} - (-110 + 4)\vec{j} + (70 + 4)\vec{k} \\ = -18\vec{i} + 106\vec{j} + 74\vec{k}$$

Again,

$$|(\vec{a} + \vec{b}) \times (\vec{a} - \vec{b})| = \sqrt{(-18)^2 + (106)^2 + (74)^2}$$

$$= \sqrt{324 + 11236 + 5476}$$

$$= \sqrt{17036}$$

① Find $\vec{a} \times \vec{b}$ if

a) $\vec{a} = 2\vec{i} - 3\vec{j}$, $\vec{b} = 2\vec{j} + 4\vec{k}$

→ Soln,

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 0 & -3 \\ 0 & 2 & 4 \end{vmatrix}$$

$$= 6\vec{i} - 8\vec{j} + 4\vec{k}$$

b) $\vec{a} = 2\vec{i} + 3\vec{j} + \vec{k}$, $\vec{b} = -\vec{i} - 2\vec{j} + 3\vec{k}$

→ Soln,

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 3 & 1 \\ -1 & -2 & 3 \end{vmatrix}$$

$$= \vec{i} - 7\vec{j} - \vec{k}$$

$$= (9+2)\vec{i} - (6+1)\vec{j} + (-4+3)\vec{k}$$

$$= 11\vec{i} - 7\vec{j} - \vec{k}$$

④ If the position vectors of the three points P, Q and R are respectively $\vec{i} + \vec{j} + 2\vec{k}$, $2\vec{i} + 3\vec{j} + \vec{k}$ and $3\vec{i} - \vec{j} + 4\vec{k}$, find a vector orthogonal to the plane PQR

→ Soln,

Given, the position vectors of three points P, Q, R are

$$\vec{OP} = \vec{i} + \vec{j} + 2\vec{k}$$

$$\vec{OQ} = 2\vec{i} + 3\vec{j} + \vec{k}$$

$$\vec{OR} = 3\vec{i} - \vec{j} + 4\vec{k}$$

We have,

$$\vec{PA} = \vec{OA} - \vec{OP} = \vec{i} + 2\vec{j} - \vec{k}$$

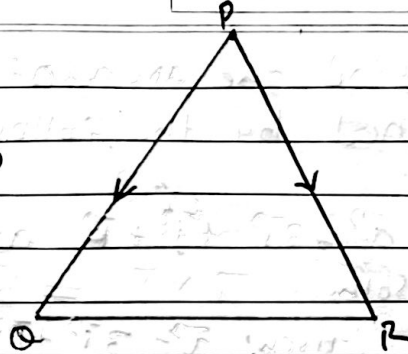
$$\vec{PR} = \vec{OR} - \vec{OP} = 2\vec{i} - 2\vec{j} + 2\vec{k}$$

The Vector perpendicular to the plane PQR is

$$\vec{PA} \times \vec{PR} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 2 & -1 \\ 2 & -2 & 2 \end{vmatrix}$$

$$= (4-2)\vec{i} - (2+2)\vec{j} + (-2-4)\vec{k}$$

$$= 2\vec{i} - 4\vec{j} - 6\vec{k}$$



5. c) Find the sine of the angle between the vectors $\vec{a} = (3, 1, 2)$ and $\vec{b} = (2, -2, 4)$.

Soln,

$$|\vec{a}| = \sqrt{3^2 + 1^2 + 2^2} = \sqrt{9 + 1 + 4} = \sqrt{14}$$

$$|\vec{b}| = \sqrt{2^2 + (-2)^2 + 4^2} = \sqrt{4 + 4 + 16} = \sqrt{24}$$

$$|\vec{a} \times \vec{b}| = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3 & 1 & 2 \\ 2 & -2 & 4 \end{vmatrix}$$

$$= \vec{i}(4+4) - \vec{j}(12-4) + \vec{k}(-6-2)$$

$$= 8\vec{i} - 8\vec{j} - 8\vec{k}$$

Now,

$$|\vec{a} \times \vec{b}| = |\vec{a}| |\vec{b}| \sin \theta$$

$$\Rightarrow \sin \theta = \frac{|\vec{a} \times \vec{b}|}{|\vec{a}| |\vec{b}|}$$

$$= \frac{8\sqrt{8^2 + (-8)^2 + (-8)^2}}{\sqrt{14} \cdot \sqrt{24}}$$

$$= \frac{8\sqrt{64 + 64 + 64}}{\sqrt{14} \cdot \sqrt{24}}$$

$$= \frac{\sqrt{192}}{\sqrt{14} \cdot \sqrt{24}}$$

$$= \frac{4\sqrt{3}}{4\sqrt{21}}$$

$$= \frac{2\sqrt{7}}{7}$$

Ans

⑧ Find the area of the triangle determined by the following pairs of vectors.

a. $\vec{a} = 3\vec{i} + \vec{j} + \vec{k}$ and $\vec{b} = \vec{i} - 2\vec{j} - \vec{k}$.

→ Soln,

Given: $\vec{a} = 3\vec{i} + \vec{j} + \vec{k}$ and $\vec{b} = \vec{i} - 2\vec{j} - \vec{k}$

We have,

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3 & 1 & 1 \\ 1 & -2 & -1 \end{vmatrix}$$

$$= (-1+2)\vec{i} - (-3-1)\vec{j} + (-6-1)\vec{k}$$

$$= \vec{i} + 4\vec{j} - 7\vec{k}$$

$$|\vec{a} \times \vec{b}| = \sqrt{1^2 + 4^2 + (-7)^2}$$

$$= \sqrt{1+16+49}$$

$$= \sqrt{66}$$

Now,

$$\text{Area of triangle} = \frac{1}{2} |\vec{a} \times \vec{b}|$$

$$= \frac{1}{2} \times \sqrt{66}$$

$$= \frac{\sqrt{66}}{2} \text{ Sq. unit}$$

⑨ Prove that the area of triangle whose vertices are $3\vec{i} - \vec{j} + 2\vec{k}$, $\vec{i} - \vec{j} - 3\vec{k}$ and $4\vec{i} - 3\vec{j} + 2\vec{k}$ as position vector is $\frac{1}{2}\sqrt{14}$

Square units.

→ Soln,

The position vectors of Vertices of a $\triangle ABC$ are

$$\vec{OA} = 3\vec{i} - \vec{j} + 2\vec{k}$$

$$\vec{OB} = \vec{i} - \vec{j} - 3\vec{k}$$

$$\vec{OC} = 4\vec{i} - 3\vec{j} + 2\vec{k}$$

We have,

$$\vec{AB} = \vec{OB} - \vec{OA}$$

$$= \vec{i} - \vec{j} - 3\vec{k} - 3\vec{i} + \vec{j} - 2\vec{k}$$

$$= -2\vec{i} - 5\vec{k}$$

$$\vec{AC} = \vec{OC} - \vec{OA}$$

$$= 4\vec{i} - 3\vec{j} + 2\vec{k} - 3\vec{i} + \vec{j} - 2\vec{k}$$

$$= \vec{i} - 2\vec{j}$$

Now,

$$\vec{AB} \times \vec{AC} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -2 & 0 & -5 \\ 1 & -2 & 0 \end{vmatrix}$$

$$= \vec{i}(0-10) - \vec{j}(0+5) + \vec{k}(4-0)$$

$$= -10\vec{i} - 5\vec{j} + 4\vec{k}$$

$$|\vec{AB} \times \vec{AC}| = \sqrt{(-10)^2 + (-5)^2 + 4^2}$$

$$= \sqrt{100 + 25 + 16}$$

$$= \sqrt{141}$$

$$\therefore \text{Area of } \triangle ABC = \frac{1}{2} |\vec{AB} \times \vec{AC}|$$

$$= \frac{1}{2} \times \sqrt{141}$$

proved

Properties of Vector Product:

① The vector product of two vectors is not commutative

$$\text{i.e. } \vec{a} \times \vec{b} \neq \vec{b} \times \vec{a}$$

But,

$$\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$$

proof

Let $\vec{a} = (a_1, a_2, a_3)$ and $\vec{b} = (b_1, b_2, b_3)$ be two vectors then,

$$\begin{aligned} \vec{a} \times \vec{b} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} \\ &= - \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ b_1 & b_2 & b_3 \\ a_1 & a_2 & a_3 \end{vmatrix} \quad (\because R_2 \leftrightarrow R_3) \\ &= -\vec{b} \times \vec{a} \end{aligned}$$

$$\text{i.e. } \vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$$

$$\therefore \vec{a} \times \vec{b} \neq \vec{b} \times \vec{a}$$

② Distributive property:

$$\vec{a} \times (\vec{b} + \vec{c}) = \vec{a} \times \vec{b} + \vec{a} \times \vec{c}$$

And;

$$(\vec{a} + \vec{b}) \times \vec{c} = \vec{a} \times \vec{c} + \vec{b} \times \vec{c}.$$

③ Associative property over scalars:

- For any two vectors $\vec{a}, \vec{b}$ and scalars m, n

$$\Rightarrow mn(\vec{a} \times \vec{b}) = (m\vec{a}) \times (n\vec{b}) = (n\vec{a}) \times (m\vec{b}) = (mn\vec{a}) \times \vec{b}$$

- ④ The vector product of two vectors $\vec{a}$ and $\vec{b}$ is perpendicular to both $\vec{a}$ and $\vec{b}$.

Proof

Let $\vec{a} = (a_1, a_2, a_3)$ and $\vec{b} = (b_1, b_2, b_3)$,
then

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$$= i(a_2 b_3 - b_2 a_3) - j(a_1 b_3 - b_1 a_3) + k(a_1 b_2 - a_2 b_1)$$

$$= (a_2 b_3 - b_2 a_3, a_3 b_1 - a_1 b_3, a_1 b_2 - a_2 b_1)$$

Now,

$$(\vec{a} \times \vec{b}) \cdot \vec{a} = (a_2 b_3 - b_2 a_3, a_3 b_1 - a_1 b_3, a_1 b_2 - a_2 b_1) \cdot (a_1, a_2, a_3)$$

$$= a_1(a_2 b_3 - a_3 b_2) + a_2(a_3 b_1 - a_1 b_3) + a_3(a_1 b_2 - a_2 b_1)$$

$$= a_1 a_2 b_3 - a_1 a_3 b_2 + a_2 a_3 b_1 - a_1 a_2 b_3 + a_1 a_3 b_2 - a_2 a_3 b_1$$

$$= 0$$

$$\text{i.e. } (\vec{a} \times \vec{b}) \cdot \vec{a} = 0$$

$$\text{i.e. } (\vec{a} \times \vec{b}) \perp \vec{a}$$

Similarly,

We can show,

$$(\vec{a} \times \vec{b}) \cdot \vec{b} = 0$$

$$\text{i.e. } (\vec{a} \times \vec{b}) \perp \vec{b}$$

Hence, the cross product $(\vec{a} \times \vec{b})$ is perpendicular to both $\vec{a}$ and $\vec{b}$.

- ⑤ For two non-zero vectors $\vec{a}$ and $\vec{b}$, $\vec{a} \times \vec{b} = 0$ if and only if they are parallel to each other.

proof.

Let $\vec{a}$ and $\vec{b}$ be two non-zero vectors and θ be the angle between them. Such that $\vec{a} \times \vec{b} = 0$.

$$\Rightarrow |\vec{a} \times \vec{b}| = 0$$

$$\Rightarrow |\vec{a}| |\vec{b}| \sin \theta = 0$$

$$\Rightarrow \sin \theta = 0 \quad (\because \text{as } \vec{a}, \vec{b} \neq 0)$$

$$\Rightarrow \theta = 0^\circ, 180^\circ$$

This proves that $\vec{a}$ and $\vec{b}$ are parallel to each other.

Conversely

Suppose; $\vec{a}$ and $\vec{b}$ are parallel to each other. Then the angle (θ) between them is either 0° or 180° .

For both cases,

$$\sin \theta = 0$$

By definition,

$$\vec{a} \times \vec{b} = |\vec{a}| |\vec{b}| \sin \theta \hat{n}$$

$$\text{i.e. } \vec{a} \times \vec{b} = 0 \quad (\because \sin \theta = 0)$$

Hence, For two non-zero vectors $\vec{a}$ and $\vec{b}$, $\vec{a} \times \vec{b} = 0$ iff they are parallel.

- ⑥ For any vector $\vec{a}$
 $\vec{a} \times \vec{a} = 0$

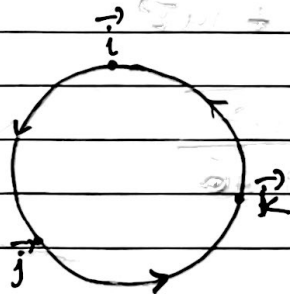
Proof

The angle θ between any vector $\vec{a}$ with itself is 0° . So, by definition.

$$\begin{aligned}\vec{a} \times \vec{a} &= |\vec{a}| |\vec{a}| \sin \theta \hat{n} \\ &= |\vec{a}| |\vec{a}| \sin 0 \hat{n} \\ &= 0\end{aligned}$$

~~###~~ Cross product of standard unit vectors:

	X	Second vector		
		$\vec{i}$	$\vec{j}$	$\vec{k}$
First vector	$\vec{i}$	0	$\vec{k}$	$-\vec{j}$
	$\vec{j}$	$-\vec{k}$	0	$\vec{i}$
	$\vec{k}$	$\vec{j}$	$-\vec{i}$	0



Q. Establish the relation between dot product and scalar product of two vectors $\vec{a}$ & $\vec{b}$.

Soln. let θ be the angle between $\vec{a}$ & $\vec{b}$.
Then.

$$\sin \theta = \frac{|\vec{a} \times \vec{b}|}{|\vec{a}| |\vec{b}|}$$

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$$

We have,

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\left(\frac{|\vec{a} \times \vec{b}|}{|\vec{a}| |\vec{b}|} \right)^2 + \left(\frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} \right)^2 = 1$$

$$\Rightarrow \frac{(\vec{a} \times \vec{b})^2}{a^2 b^2} + \frac{(\vec{a} \cdot \vec{b})^2}{a^2 b^2} = 1$$

$$\Rightarrow (\vec{a} \times \vec{b})^2 + (\vec{a} \cdot \vec{b})^2 = a^2 b^2$$

③ Find the Vector and the unit vector orthogonal to each of the following pair of vector.

a) $\vec{a} = 4\vec{i} - 2\vec{j} + 3\vec{k}$ and $\vec{b} = 5\vec{i} + \vec{j} - 4\vec{k}$

→ Soln.

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 4 & -2 & 3 \\ 5 & 1 & -4 \end{vmatrix}$$

$$= (8-3)\vec{i} - (-16-15)\vec{j} + (4+10)\vec{k}$$

$$= 5\vec{i} + 31\vec{j} + 14\vec{k}$$

$$\begin{aligned} |\vec{a} \times \vec{b}| &= \sqrt{5^2 + 31^2 + 14^2} \\ &= \sqrt{25 + 961 + 196} \\ &= \sqrt{1182} \\ &= \cancel{33.64} \end{aligned}$$

Now,

$$\begin{aligned} \frac{(\vec{a} \times \vec{b})}{|\vec{a} \times \vec{b}|} &= \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|} \\ &= \frac{5\vec{i} + 31\vec{j} + 14\vec{k}}{\sqrt{1182}} \end{aligned}$$

b) $\vec{a} = 6\vec{i} + 3\vec{j} - 5\vec{k}$ and $\vec{b} = \vec{i} - 4\vec{j} + 2\vec{k}$

→ Soln,

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 6 & 3 & -5 \\ 1 & -4 & 2 \end{vmatrix}$$

$$= i(6-20) - j(12+5) + k(-24-3)$$

$$= -14\vec{i} - 17\vec{j} - 27\vec{k}$$

$$|\vec{a} \times \vec{b}| = \sqrt{(-14)^2 + (-17)^2 + (-27)^2}$$

$$= \sqrt{196 + 289 + 729}$$

$$= \sqrt{1214}$$

Now,

$$(\hat{\vec{a} \times \vec{b}}) = \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|}$$

$$= \frac{-14\vec{i} - 17\vec{j} - 27\vec{k}}{\sqrt{1214}}$$

⑤

a) If θ be the angle between the vectors $\vec{a} = 2\vec{i} - 3\vec{j} + 5\vec{k}$ and $\vec{b} = 4\vec{i} - 7\vec{k}$, find the value of $\sin \theta$.

→ Soln,

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -3 & 5 \\ 4 & 0 & -7 \end{vmatrix}$$

$$= (21-0)\vec{i} - (-14-20)\vec{j} + (0+12)\vec{k}$$

$$= 21\vec{i} + 34\vec{j} + 12\vec{k}$$

$$|\vec{a} \times \vec{b}| = \sqrt{21^2 + 34^2 + 12^2}$$

$$= \sqrt{1741}$$

Now,

$$|\vec{a}| = \sqrt{2^2 + (-3)^2 + 5^2}$$

$$= \sqrt{4 + 9 + 25}$$

$$= \sqrt{38}$$

$$\begin{aligned}
 |\vec{b}| &= \sqrt{4^2 + 0^2 + (-7)^2} \\
 &= \sqrt{16 + 49} \\
 &= \sqrt{65}
 \end{aligned}$$

Again,

$$\begin{aligned}
 |\vec{a} \times \vec{b}| &= |\vec{a}| |\vec{b}| \sin \theta \\
 \Rightarrow \sqrt{1741} &= \sqrt{38} \times \sqrt{65} \sin \theta \\
 \Rightarrow \sin \theta &= \frac{\sqrt{1741}}{\sqrt{38} \times \sqrt{65}} \\
 &= \frac{\sqrt{1741}}{\sqrt{2470}} \\
 &= \frac{\sqrt{1741}}{\sqrt{2470}}
 \end{aligned}$$

b) Find the sine of the angle between the vectors $\vec{a} = 3\vec{i} + \vec{j} + 2\vec{k}$ and $\vec{b} = 2\vec{i} - 2\vec{j} + 9\vec{k}$

✓ Soln,

$$\begin{aligned}
 \vec{a} &= 3\vec{i} + \vec{j} + 2\vec{k} \\
 |\vec{a}| &= \sqrt{3^2 + 1^2 + 2^2} \\
 &= \sqrt{9 + 1 + 4} \\
 &= \sqrt{14}
 \end{aligned}$$

$$\begin{aligned}
 \vec{b} &= 2\vec{i} - 2\vec{j} + 9\vec{k} \\
 |\vec{b}| &= \sqrt{2^2 + (-2)^2 + 9^2} \\
 &= \sqrt{4 + 4 + 81} \\
 &= \sqrt{89}
 \end{aligned}$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3 & 1 & 2 \\ 2 & -2 & 9 \end{vmatrix}$$

$$\begin{aligned}
 &= (9 + 4)\vec{i} - (27 - 4)\vec{j} + (-6 - 2)\vec{k} \\
 &= 13\vec{i} - 23\vec{j} - 8\vec{k}
 \end{aligned}$$

$$|\vec{a} \times \vec{b}| = \sqrt{13^2 + (-23)^2 + (-8)^2}$$

$$= \sqrt{762}$$

$$|\vec{a} \times \vec{b}| = |\vec{a}| |\vec{b}| \sin \theta$$

$$\Rightarrow \sqrt{762} = \sqrt{14} \times \sqrt{89} \sin \theta$$

$$\Rightarrow \sin \theta = \frac{\sqrt{762}}{\sqrt{14} \times \sqrt{89}}$$

$$= \frac{\sqrt{762}}{\sqrt{1246}}$$

$$= \frac{\sqrt{381}}{\sqrt{623}}$$

#.

- ⑥ Find the area of the parallelogram whose two adjacent sides are determined by the following pairs of vectors.

a) $\vec{a} = 7\vec{i} + 8\vec{j} - \vec{k}$ and $\vec{b} = 10\vec{i} - 11\vec{j} + 12\vec{k}$

→ Soln,

~~Area~~ Area of the parallelogram having adjacent sides $\vec{a}$ and $\vec{b}$ is given by $|\vec{a} \times \vec{b}|$.

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 7 & 8 & -1 \\ 10 & -11 & 12 \end{vmatrix}$$

$$= \vec{i}(96 - 11) - \vec{j}(84 + 10) + \vec{k}(-77 - 80)$$

$$= 85\vec{i} - 94\vec{j} - 157\vec{k}$$

Now,

$$|\vec{a} \times \vec{b}| = \sqrt{85^2 + (-94)^2 + (-157)^2}$$

$$= \sqrt{40710}$$

$$\therefore \text{Area of parallelogram} = \sqrt{40710} \text{ sq. units.}$$

⑥ $\vec{a} = \vec{i} + 2\vec{j} + 3\vec{k}$ and $\vec{b} = \vec{i} - 2\vec{j} + 4\vec{k}$

→ Soln,

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 2 & 3 \\ 1 & -2 & 4 \end{vmatrix}$$

$$= (8+6)\vec{i} - (4-3)\vec{j} + (-2-2)\vec{k}$$

$$= 14\vec{i} - 1\vec{j} - 4\vec{k}$$

$$|\vec{a} \times \vec{b}| = \sqrt{14^2 + (-1)^2 + (-4)^2}$$

$$= \sqrt{196 + 1 + 16}$$

$$= \sqrt{213}$$

∴ Area of parallelogram = $\sqrt{213}$ sq. units

⑦ Find the area of the parallelogram whose diagonals are represented by the vectors $\vec{i} + \vec{j} - \vec{k}$ and $\vec{i} - \vec{j} + \vec{k}$.

→ Soln,

let $\vec{a}$ and $\vec{b}$ be two diagonals of a parallelogram.

i.e. $\vec{a} = \vec{i} + \vec{j} - \vec{k}$

$\vec{b} = \vec{i} - \vec{j} + \vec{k}$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 1 & -1 \\ 1 & -1 & 1 \end{vmatrix}$$

$$= (1-1)\vec{i} - \vec{j}(1+1) + \vec{k}(-1-1)$$

$$= -2\vec{j} - 2\vec{k}$$

$$|\vec{a} \times \vec{b}| = \sqrt{(-2)^2 + (-2)^2}$$

$$= \sqrt{4+4}$$

$$= \sqrt{8}$$

Area of parallelogram with diagonals

$$= \frac{1}{2} |\vec{a}' \times \vec{b}'|$$

$$= \frac{1}{2} \times \sqrt{8}$$

$$= \sqrt{2} \text{ Sq. units.}$$

8.

b) $\vec{a}' = 3\vec{i}' + 4\vec{j}'$ and $\vec{b}' = -5\vec{i}' + 7\vec{j}'$

→ Soln,

$$\vec{a}' \times \vec{b}' = \begin{vmatrix} \vec{i}' & \vec{j}' & \vec{k}' \\ 3 & 4 & 0 \\ -5 & 7 & 0 \end{vmatrix}$$

$$= 0\vec{i}' - 0\vec{j}' + (21 + 20)\vec{k}'$$

$$= 41\vec{k}'$$

$$|\vec{a}' \times \vec{b}'| = \sqrt{41^2} = 41$$

Now,

$$\text{Area of triangle} = \frac{1}{2} |\vec{a}' \times \vec{b}'|$$

$$= \frac{41}{2} \text{ Sq. units.}$$

(10)

a) If $|\vec{a}'| = 15$, $|\vec{b}'| = 4$ and $|\vec{a}' \times \vec{b}'| = 36$, find the value of $\vec{a}' \cdot \vec{b}'$.

→ Soln,

$$|\vec{a}' \times \vec{b}'| = |\vec{a}'| |\vec{b}'| \sin \theta$$

$$\Rightarrow 36 = 15 \times 4 \sin \theta$$

$$\Rightarrow \frac{36}{60} = \sin \theta$$

$$\Rightarrow \sin \theta = \frac{3}{5}$$

$$\theta = \sin^{-1}\left(\frac{3}{5}\right) = 36.86$$

Now,

$$\begin{aligned}\vec{a} \cdot \vec{b} &= |\vec{a}| \cdot |\vec{b}| \cos \theta \\ &= 15 \times 4 \times \cos 36.86 \\ &= \cancel{30} \times 4 \times \frac{4}{5} \\ &= 48\end{aligned}$$

b) Given $|\vec{a}| = 9$, $|\vec{b}| = 5$ and $\vec{a} \cdot \vec{b} = 36$, find $|\vec{a} \times \vec{b}|$.

→ Soln,

$$(\vec{a} \times \vec{b})^2 + (\vec{a} \cdot \vec{b})^2 = a^2 b^2$$

$$\Rightarrow |\vec{a} \times \vec{b}|^2 + (36)^2 = 9^2 \times 5^2$$

$$\Rightarrow |\vec{a} \times \vec{b}|^2 + 1296 = 81 \times 25$$

$$\Rightarrow |\vec{a} \times \vec{b}|^2 = 2025 - 1296$$

$$\Rightarrow |\vec{a} \times \vec{b}|^2 = 729$$

$$\Rightarrow |\vec{a} \times \vec{b}| = \sqrt{729}$$

$$\Rightarrow |\vec{a} \times \vec{b}| = 27$$

$$\therefore |\vec{a} \times \vec{b}| = 27$$

c) If $\vec{a} = \vec{i} + \vec{j} + \vec{k}$, $\vec{c} = \vec{j} - \vec{j}$, then find $\vec{b}$ such that $\vec{a} \times \vec{b} = \vec{c}$ and $\vec{a} \cdot \vec{b} = 3$

→ Given,

$$\vec{a} = \vec{i} + \vec{j} + \vec{k}$$

$$\vec{c} = \vec{j} - \vec{j}$$

$$\vec{a} \times \vec{b} = \vec{c}$$

$$\vec{a} \cdot \vec{b} = 3$$

$$\text{Let } \vec{b} = (b_1, b_2, b_3)$$

Now,

$\vec{a} \times \vec{b} =$	$\vec{i}$	$\vec{j}$	$\vec{k}$
	1	1	1
	b_1	b_2	b_3

$$= (b_3 - b_2)\vec{i}' - (b_3 - b_1)\vec{j}' + (b_2 - b_1)\vec{k}'$$

Then,

$$\vec{a}' \times \vec{b}' = \vec{c}'$$

$$(b_3 - b_2, b_1 - b_3, b_2 - b_1) = \lambda (1, -1, 0)$$

Equating L.H.S and R.H.S,

$$b_3 - b_2 = 1 \rightarrow (i) \Rightarrow b_3 = 1 + b_2$$

$$b_1 - b_3 = -1 \rightarrow (ii) \Rightarrow b_1 = b_3 - 1$$

$$b_2 - b_1 = 0 \rightarrow (iii) \Rightarrow b_2 = b_1$$

Again,

$$\vec{a}' \cdot \vec{b}' = 3$$

$$(\vec{i}' + \vec{j}' + \vec{k}') \cdot (b_1\vec{i}' + b_2\vec{j}' + b_3\vec{k}') = 3$$

$$b_1 + b_2 + b_3 = 3$$

$$b_1 + b_1 + b_1 + 1 = 3 \quad (\text{from (i) and (iii)})$$

$$3b_1 = 3 - 1$$

$$3b_1 = 2$$

$$b_1 = \frac{2}{3}$$

$$b_1 = b_2 = \frac{2}{3}$$

$$b_3 = 1 + b_1 = \frac{2}{3} + 1 = \frac{5}{3}$$

$$\therefore \vec{b}' = \left(\frac{2}{3}, \frac{2}{3}, \frac{5}{3} \right) = \frac{1}{3} (2\vec{i}' + 2\vec{j}' + 5\vec{k}')$$

e) If $\vec{a}' + \vec{b}' + \vec{c}' = 0$, prove that $\vec{a}' \times \vec{b}' = \vec{b}' \times \vec{c}' = \vec{c}' \times \vec{a}'$.

→ Given,

$$\vec{a}' + \vec{b}' + \vec{c}' = 0 \rightarrow \textcircled{1}$$

Taking cross product on both sides of eqn $\textcircled{1}$ by $\vec{a}'$.

$$\vec{a}' \times (\vec{a}' + \vec{b}' + \vec{c}') = \vec{a}' \times \vec{0}$$

$$\Rightarrow \vec{a}' \times \vec{a}' + \vec{a}' \times \vec{b}' + \vec{a}' \times \vec{c}' = \vec{0}$$

$$\Rightarrow 0 + \vec{a}' \times \vec{b}' + \vec{a}' \times \vec{c}' = \vec{0}$$

$$\Rightarrow \vec{a}' \times \vec{b}' - \vec{c}' \times \vec{a}' = \vec{0}$$

$$\Rightarrow \vec{a}' \times \vec{b}' = \vec{c}' \times \vec{a}' \rightarrow (2)$$

Again, taking cross product on both sides of eqn (i) by $\vec{b}'$,

$$\vec{b}' \times (\vec{a}' + \vec{b}' + \vec{c}') = \vec{b}' \times \vec{0}$$

$$\Rightarrow \vec{b}' \times \vec{a}' + \vec{b}' \times \vec{b}' + \vec{b}' \times \vec{c}' = \vec{0}$$

$$\Rightarrow -\vec{a}' \times \vec{b}' + 0 + \vec{b}' \times \vec{c}' = \vec{0}$$

$$\Rightarrow \vec{a}' \times \vec{b}' = \vec{b}' \times \vec{c}' \rightarrow (3)$$

Combining eqn (2) and (3) we get.

$$\vec{a}' \times \vec{b}' = \vec{b}' \times \vec{c}' = \vec{c}' \times \vec{a}'$$

Proved

- (ii) If $\vec{a}', \vec{b}', \vec{c}', \vec{d}'$ are the position vector of the vertices of a quadrilateral ABCD, prove that the vector area of the quadrilateral ABCD is $\frac{1}{2} [\vec{a}' \times \vec{b}' + \vec{b}' \times \vec{c}' + \vec{c}' \times \vec{d}' + \vec{d}' \times \vec{a}']$.

→ Soln.

(Given: The position vectors of vertices of a quadrilateral ABCD are

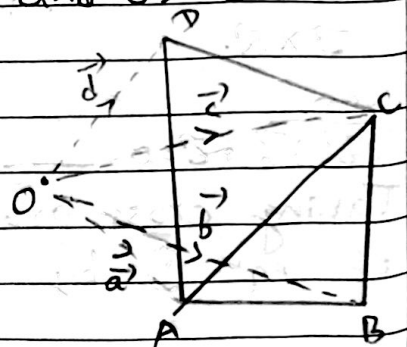
$$\vec{OA}' = \vec{a}', \vec{OB}' = \vec{b}', \vec{OC}' = \vec{c}' \text{ and } \vec{OD}' = \vec{d}'.$$

We have,

$$\vec{AB}' = \vec{OB}' - \vec{OA}' = \vec{b}' - \vec{a}'$$

$$\vec{AC}' = \vec{OC}' - \vec{OA}' = \vec{c}' - \vec{a}'$$

$$\vec{AD}' = \vec{OD}' - \vec{OA}' = \vec{d}' - \vec{a}'$$



Here,

Vector area of quadrilateral ABCD
= Vector area of $\triangle ABC$ + Vector area of $\triangle ACD$

$$= \frac{1}{2} (\vec{AB} \times \vec{AC}) + \frac{1}{2} (\vec{AC} \times \vec{AD})$$

$$= \frac{1}{2} [(\vec{b} - \vec{a}) \times (\vec{c} - \vec{a}) + \vec{c} - \vec{a} \times (\vec{d} - \vec{a})]$$

$$= \frac{1}{2} [\vec{b} \times \vec{c} - \vec{b} \times \vec{a} - \vec{a} \times \vec{c} + \vec{a} \times \vec{a} + \vec{c} \times \vec{d} - \vec{c} \times \vec{a} - \vec{a} \times \vec{d} + \vec{a} \times \vec{a}]$$

$$= \frac{1}{2} [\vec{b} \times \vec{c} + \vec{a} \times \vec{b} - \cancel{\vec{a} \times \vec{c}} + 0 + \vec{c} \times \vec{d} + \cancel{\vec{a} \times \vec{c}} + \cancel{\vec{a} \times \vec{d}}]$$

$$= \frac{1}{2} [\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{d} + \vec{d} \times \vec{a}]$$

Proved

10) d) Given any three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$,
prove that $\vec{a} \times (\vec{b} + \vec{c}) + \vec{b} \times (\vec{c} + \vec{a}) + \vec{c} \times (\vec{a} + \vec{b}) = 0$

$\rightarrow$ Soln,

$$\begin{aligned} & \vec{a} \times (\vec{b} + \vec{c}) + \vec{b} \times (\vec{c} + \vec{a}) + \vec{c} \times (\vec{a} + \vec{b}) \\ &= \vec{a} \times \vec{b} + \vec{a} \times \vec{c} + \vec{b} \times \vec{c} + \vec{b} \times \vec{a} + \vec{c} \times \vec{a} + \vec{c} \times \vec{b} \\ &= \cancel{\vec{a} \times \vec{b}} + \cancel{\vec{a} \times \vec{c}} + \vec{b} \times \vec{c} - \cancel{\vec{a} \times \vec{b}} - \cancel{\vec{a} \times \vec{c}} - \cancel{\vec{b} \times \vec{c}} \\ &= 0 \end{aligned}$$

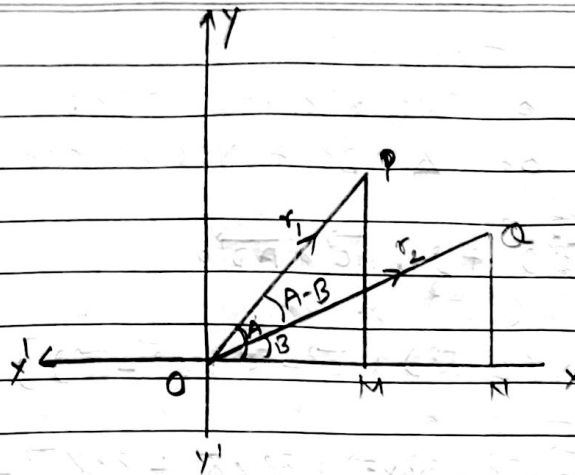
Proved

prove by vector method:

$$\sin(A-B) = \sin A \cdot \cos B - \cos A \cdot \sin B$$

Proof

Let xox' and yoy' respectively denotes the x and y -axes. Suppose P and Q be two points such that $\angle Pox = A$. Similarly



$\angle QOx = B$, $OP = r_1$, & $OQ = r_2$, then $\angle POQ = A - B$
 Draw PM and QN perpendicular to x -axis.

In right angled $\triangle POM$,

$$\sin A = \frac{PM}{r_1} \Rightarrow PM = r_1 \sin A$$

$$\cos A = \frac{OM}{r_1} \Rightarrow OM = r_1 \cos A$$

So,

coordinates of $P = (OM, PM) = (r_1 \cos A, r_1 \sin A)$

$$\therefore \vec{OP} = (r_1 \cos A, r_1 \sin A, 0)$$

Similarly,

coordinates of $Q = (r_2 \cos B, r_2 \sin B)$

$$\therefore \vec{OQ} = (r_2 \cos B, r_2 \sin B, 0)$$

Now,

$$\vec{OP} \times \vec{OQ} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ r_1 \cos A & r_1 \sin A & 0 \\ r_2 \cos B & r_2 \sin B & 0 \end{vmatrix}$$

$$= 0\vec{i} + 0\vec{j} + (r_1 \cos A \cdot r_2 \sin B - r_1 r_2 \sin A \cos B) \vec{k}$$

$$= 0\vec{i} + 0\vec{j} - r_1 r_2 (\sin A \cos B - \cos A \sin B) \vec{k}$$

$$\begin{aligned}
 |\vec{OP'} \times \vec{OA}| &= \sqrt{0^2 + 0^2 + (-r_1 r_2 (\sin A \cos B - \cos A \sin B))^2} \\
 &= \sqrt{r_1^2 r_2^2 (\sin A \cos B - \cos A \sin B)^2} \\
 &= r_1 r_2 (\sin A \cos B - \cos A \sin B)
 \end{aligned}$$

Now,

$$\begin{aligned}
 |\vec{OP'} \times \vec{OA}| &= |\vec{OP'}| |\vec{OA}| \sin \theta (A-B) \\
 r_1 r_2 (\sin A \cos B - \cos A \sin B) &= r_1 r_2 \sin(A-B)
 \end{aligned}$$

$$\therefore \sin(A-B) = \sin A \cos B - \cos A \sin B$$

Proved.

12) In any $\triangle ABC$, prove by vector method

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

proof

Let ABC be a triangle
 such that $\vec{AB} = \vec{c}$, $\vec{BC} = \vec{a}$,
 $\vec{AC} = \vec{b}$

Now,

By triangle law,

$$\vec{a} + \vec{b} + \vec{c} = 0 \rightarrow (i)$$

Taking cross product on both sides of eqn (i)
 by $\vec{a}$

$$\vec{a} \times (\vec{a} + \vec{b} + \vec{c}) = \vec{a} \times 0$$

$$\Rightarrow \vec{a} \times \vec{a} + \vec{a} \times \vec{b} + \vec{a} \times \vec{c} = 0$$

$$\Rightarrow 0 + \vec{a} \times \vec{b} + -\vec{c} \times \vec{a} = 0$$

$$\Rightarrow \vec{a} \times \vec{b} = \vec{c} \times \vec{a} \rightarrow (ii)$$

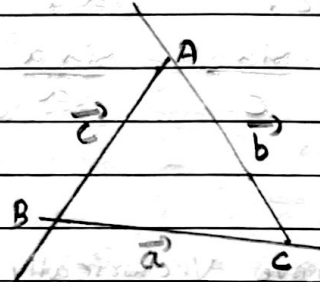
Again, taking cross product on both sides of
 eqn (i) by $\vec{b}$

$$\vec{b} \times (\vec{a} + \vec{b} + \vec{c}) = 0$$

$$\Rightarrow \vec{b} \times \vec{a} + 0 + \vec{b} \times \vec{c} = 0$$

$$\Rightarrow -\vec{a} \times \vec{b} + \vec{b} \times \vec{c} = 0$$

$$\Rightarrow \vec{a} \times \vec{b} = \vec{b} \times \vec{c} \rightarrow (iii)$$



Combining (ii) & (iii) we get.

$$\vec{a}' \times \vec{b}' = \vec{b}' \times \vec{c}' = \vec{c}' \times \vec{a}'$$

Now,

angle between $\vec{a}'$ & $\vec{b}'$ is $(\pi - c)$, $\vec{b}'$ & $\vec{c}'$ is $(\pi - a)$ & $\vec{c}'$ and $\vec{a}'$ is $(\pi - b)$

Now,

$$|\vec{a}'||\vec{b}'| \cdot \sin(\pi - c) = |\vec{b}'||\vec{c}'| \cdot \sin(\pi - a) = |\vec{c}'||\vec{a}'| \cdot \sin(\pi - b)$$

$$\Rightarrow |\vec{a}'||\vec{b}'| \cdot \sin c = |\vec{b}'||\vec{c}'| \cdot \sin a = |\vec{c}'||\vec{a}'| \cdot \sin b$$

$$\Rightarrow \frac{|\vec{a}'||\vec{b}'| \sin c}{|\vec{a}'||\vec{b}'||\vec{c}'|} = \frac{|\vec{b}'||\vec{c}'| \sin a}{|\vec{a}'||\vec{b}'||\vec{c}'|} = \frac{|\vec{c}'||\vec{a}'| \sin b}{|\vec{a}'||\vec{b}'||\vec{c}'|}$$

$$\Rightarrow \frac{\sin c}{|\vec{c}'|} = \frac{\sin a}{|\vec{a}'|} = \frac{\sin b}{|\vec{b}'|}$$

$$\therefore \frac{\sin c}{c} = \frac{\sin a}{a} = \frac{\sin b}{b} \quad [\because |\vec{c}'| = c]$$

proved

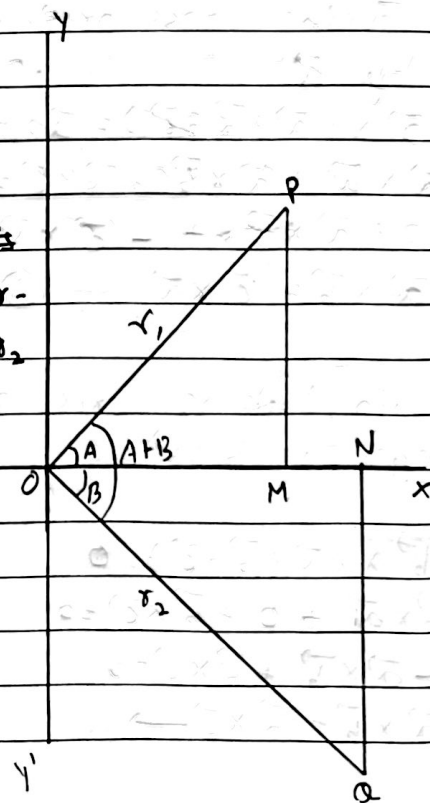
13. Prove Vectorically that:

$$a) \sin(A+B) = \sin A \cdot \cos B + \cos A \cdot \sin B$$

Let XOX' and YOY' respectively denotes the x -axis and y -axis. Suppose P and Q be two points ~~on x -axis~~ such that $\angle POX = A$. Similarly $\angle QOX = B$, $OP = r_1$ & $OQ = r_2$ then $\angle POQ = A+B$.

Draw PM and QN perpendicular to x -axis

In right angled $\triangle POM$,



$$\sin A = \frac{PM}{r_1} \Rightarrow PM = r_1 \sin A$$

$$\cos A = \frac{OM}{r_1} \Rightarrow OM = r_1 \cos A$$

So,

coordinates of $P = (OM, PM) = (r_1 \cos A, r_1 \sin A)$

$$\therefore \vec{OP} = (r_1 \cos A, r_1 \sin A, 0)$$

similarly

coordinates of $Q = (r_2 \cos B, -r_2 \sin B)$

negative sign in y-axis because point lies on 4th quadrant.

$$\therefore \vec{OQ} = (r_2 \cos B, -r_2 \sin B, 0)$$

Now,

$$\vec{OQ} \times \vec{OP} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ r_2 \cos B & -r_2 \sin B & 0 \\ r_1 \cos A & r_1 \sin A & 0 \end{vmatrix}$$

$$= 0\vec{i} + 0\vec{j} + (r_1 r_2 \sin A \cos B + r_1 r_2 \sin B \cos A)\vec{k}$$

$$\begin{aligned} |\vec{OQ} \times \vec{OP}| &= \sqrt{0^2 + 0^2 + (r_1 r_2 (\sin A \cos B + \cos A \sin B))^2} \\ &= \sqrt{r_1^2 r_2^2 (\sin A \cos B + \cos A \sin B)^2} \\ &= r_1 r_2 (\sin A \cos B + \cos A \sin B) \end{aligned}$$

$$|\vec{OQ} \times \vec{OP}| = |\vec{OQ}| |\vec{OP}| \sin(A+B)$$

$$r_1 r_2 (\sin A \cos B + \cos A \sin B) = r_1 r_2 \sin(A+B)$$

$$\Rightarrow \therefore \sin(A+B) = \sin A \cos B + \cos A \sin B$$

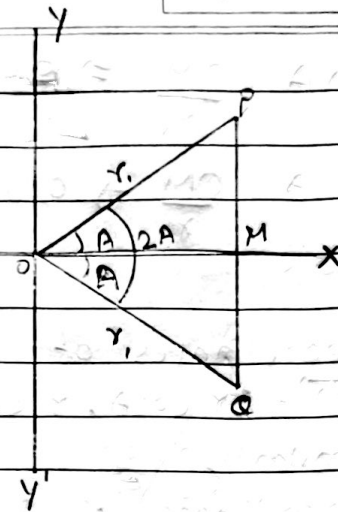
$$b) \sin 2A = 2 \sin A \cos A$$

Let xox' and yoy' respectively denotes the x-axis and y-axis. Suppose P and Q be two points such that $\angle POX = A$. Similarly

$\angle POX = A$, $OP = OQ = r$, then

$\angle POQ = 2A$.

Draw PM and QM perpendicular to x -axis.



In right angled $\triangle POM$

$$\sin A = \frac{PM}{OP} \Rightarrow PM = r \sin A$$

$$\cos A = \frac{OM}{r} \Rightarrow OM = r \cos A$$

So,

Coordinates of $P = (OM, PM) = (r \cos A, r \sin A)$

$$\therefore \vec{OP} = (r \cos A, r \sin A)$$

Similarly,

coordinates of $Q = (r \cos A, -r \sin A)$

negative sign because Q lies on 4th quadrant

$$\therefore \vec{OQ} = (r \cos A, -r \sin A)$$

Now,

$$\begin{aligned} \vec{OQ} \times \vec{OP} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ r \cos A & -r \sin A & 0 \\ r \cos A & r \sin A & 0 \end{vmatrix} \\ &= 0\hat{i} + 0\hat{j} + (r^2 \sin A \cos A + r^2 \sin A \cos A)\hat{k} \end{aligned}$$

$$\begin{aligned} |\vec{OQ} \times \vec{OP}| &= \sqrt{0^2 + 0^2 + (r^2 (\sin A \cos A + \sin A \cos A))^2} \\ &= \sqrt{(r^2)^2 (2 \sin A \cos A)^2} \\ &= r^2 (2 \sin A \cos A) \end{aligned}$$

$$\begin{aligned} \sin(2A) &= \frac{|\vec{OQ} \times \vec{OP}|}{|\vec{OQ}| |\vec{OP}|} \\ &= \frac{r^2 (2 \sin A \cos A)}{r \cdot r} \end{aligned}$$

$$\sin 2A = 2 \sin A \cos A$$

Prove